Review Article

Management of Asymptomatic Cholelithiasis

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Abstract

Asymptomatic gallstones are found in approximately 10%–20% of the adult population, with only 1%–4% developing symptoms annually. Although complications such as biliary colic, acute cholecystitis or pancreatitis may occur, the overall risk remains low. Laparoscopic cholecystectomy, though effective with a >95% success rate, carries a morbidity of ~2%–6% and a mortality risk of <0.5%, making routine surgery unwarranted in most asymptomatic individuals. Observation is recommended in most cases. Prophylactic cholecystectomy is considered for high-risk groups, including patients with gallstones >2–3 cm, those with sickle cell disease or candidates for bariatric or organ transplant surgery. While the risk of gall bladder cancer in asymptomatic cholelithiasis is <0.3%, enhanced surveillance is advised in regions with high cancer prevalence. Routine cholecystectomy for cancer prevention is not supported by current evidence.

Keywords: Cholelithiasis, gallstone, management

Introduction

Asymptomatic gallstone disease, also known as silent cholelithiasis, refers to the presence of gallstones in the gallbladder without any associated symptoms. It is a common condition worldwide, affecting approximately 10%–20% of the adult population, with prevalence rates reaching up to 30%–40% in certain high-risk regions such as South America, Northern India and among Native American populations.[1,2] The burden is expected to rise further due to increasing rates of obesity, sedentary lifestyles and metabolic syndrome.

The widespread use of ultrasonography – during pregnancy, evaluation of abdominal complaints and routine health screenings such as pre-employment checkups – has led to the incidental detection of asymptomatic gallstones in an increasing number of otherwise healthy individuals.

Numerous studies have explored the prevalence, natural history, risk factors and management strategies for silent gallstones. However, despite extensive research, the clinical approach to asymptomatic gallstones remains controversial and varies significantly across different regions. This chapter will provide an overview of global data on asymptomatic gallstones and summarise current perspectives on their management.

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Prevalence

Gallstone disease is a common condition, with a reported prevalence ranging from 10% to 20%.[1,2] Notably, more than 80% of affected individuals remain asymptomatic.[3] The annual rate of progression from asymptomatic to symptomatic gallstone disease is estimated at 1%–2%.[4]

The overall prevalence of gallstones differs by region, estimated at 5%–10% in Eastern countries and 10%–20% in Western populations.[5] Country-specific prevalence includes: 10%–15% in the United States, 14.5% in Mexico, 15% in Sweden, 5.9% in Italy, 20% in Taiwan, 4%–10% in India, 6% in Bangladesh, 3.5% in Thailand and 3.2% in Japan.[5-7] In the United States, the prevalence of asymptomatic gallstones among individuals over 40 years of age is 11% in men and 14% in women.[8]

It is important to interpret these prevalence rates with caution, as the studies cited used varying diagnostic methods and

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included different age groups and populations. Nonetheless, they provide a general overview of the global burden and geographic variation of asymptomatic gallstone disease.

Natural History of Asymptomatic Gallstone Disease

The natural history of asymptomatic gallstone disease has been investigated across several large-scale, longitudinal studies. Among faculty members at the University of Michigan, 113 individuals with asymptomatic gallstones were followed prospectively at intervals of 5, 10 and 15 years. The cumulative incidence of developing symptoms was 10%, 15% and 18%, respectively, indicating a low but progressive risk over time.[9] Importantly, all individuals who developed complications – such as acute calculus cholecystitis, cholangitis, pancreatitis or gallstone ileus – were first preceded by an episode of biliary colic, suggesting that pain often serves as an early warning signal for more severe events.

A similar trend was observed in a 25-year longitudinal study from the Bay Area in San Francisco, which followed 123 patients with asymptomatic gallstones. The average annual incidence of symptoms or complications was approximately 1%, reinforcing the slow progression of disease in most individuals.[10]

In a large population-based cohort study from Italy involving 11,229 subjects with gallstones, 73.1% were initially asymptomatic. After a mean follow-up of 8.7 years, only 22% of asymptomatic patients developed symptoms, indicating a relatively low cumulative incidence.[11] Likewise, a retrospective cohort study of 22,257 asymptomatic gallstone patients reported that 14.5% became symptomatic over a median follow-up period of 4.6 years.[4]

In the paediatric population, the natural history appears somewhat different. A spontaneous resolution rate of 16.5% has been documented among infants and children with asymptomatic gallstones, suggesting a more benign course in this group.[12]

When analysed longitudinally, these data show that the cumulative incidence of symptoms ranges from 9.9% to 28% over 6–25 years, with an annual conversion rate generally <2%.[9-11] Risk stratification based on symptomatology and time since diagnosis reveals that the longer a patient remains asymptomatic, the less likely they are to develop symptoms in the future. Furthermore, it is rare for complications to occur without a prior episode of biliary colic, which serves as a clinically useful predictor of progression.[9]

These findings support a conservative approach in most asymptomatic individuals. The decision to intervene should be individualised, considering patient-specific risk factors such as age, comorbidities, stone characteristics (e.g. size > 2–3 cm), anatomical variants and personal preferences.

Asymptomatic Gallstone And Gallbladder Cancer

The association between gallstones and gall bladder cancer remains controversial. Although the prevalence of gallstones is high in countries such as the USA, the United Kingdom and much of Western Europe, the incidence of gall bladder cancer in these regions is relatively low.[13,14] Even in regions with a high incidence of gall bladder cancer, such as India and Japan, not all cases are associated with gallstones – 20%–30% of gall bladder cancer cases in India and 40%–50% in Japan occur without gallstones.[7,15,16]

Gallstones may contribute to carcinogenesis through chronic irritation and inflammation, which can lead to the local production of carcinogens such as secondary bile acids. This has led to the hypothesis of a dysplasia–carcinoma sequence. However, this theory is unproven. The dysplasia–carcinoma pathway typically results in squamous carcinoma, whereas 90% of gallbladder cancers are adenocarcinomas.[17] Bile acids have also been implicated in the pathogenesis of duodenal adenocarcinoma, warranting further research into their role in gall bladder carcinogenesis.

While the absolute risk of gall bladder cancer in individuals with asymptomatic gallstones is low, it is not negligible. The risk varies depending on several factors including stone size and volume, duration of gallstone disease, patient age and comorbidities. A study covering 40 international cancer registries reported that 10%–15% of adults have gallstones, but only 0.3%–3.0% of these individuals develop gall bladder carcinoma over a 20-year period.[18]

Stone size is a significant risk factor. Gallstones larger than 3 cm are associated with a 10-fold increased risk of gall bladder cancer compared to those smaller than 1 cm.[19,20] Similarly, stone volume has also been linked to cancer risk. Gallstone volumes of 6 mL, 8 mL and 10 mL correspond to a ×5, ×7 and ×11 relative risk of gall bladder carcinoma, respectively.[21]

The duration of gallstone disease is another key factor. In a Chilean case-control study of 228 individuals, 15% of patients with gall bladder carcinoma had a long-standing history of gallstones compared to 4% in the control group, yielding an odds ratio of 11 (confidence interval 1.4–85.2).[22] This may explain the high incidence of gall bladder carcinoma in North Indian women, where gallstone disease often begins at a young age and remains undiagnosed for years.

The composition of gallstones may also influence cancer risk. In North India, where gall bladder carcinoma rates are high, over 80% of gallstones are cholesterol rich, whereas in South India, where cancer rates are lower, pigment stones are more common.

Genetic, environmental and infectious factors also play important roles. The risk of gall bladder cancer may be higher in individuals with a family history of the disease, genetic mutations or exposure to environmental toxins. In India,

factors such as arsenic-contaminated water, pesticides and specific dietary habits have been implicated in gall bladder carcinogenesis.[7,23,24] Supporting the role of genetics, a study of Indians living in the USA found a higher incidence of gall bladder cancer than in Native Americans, suggesting a hereditary predisposition.[25]

Infectious and inflammatory conditions, such as chronic Salmonella infection or primary sclerosing cholangitis, have also been associated with increased risk.[26]

In East Asian countries like Korea and Japan, where gall bladder cancer incidence is high, the predominant risk factor is not gallstones but rather congenital biliary anomalies, particularly pancreaticobiliary maljunction.[27]

In summary, while the risk of developing gall bladder cancer from asymptomatic gallstones is not zero, it is still relatively low. However, individuals with asymptomatic gallstones should still be monitored closely by a healthcare professional to detect any changes that may arise. This is more relevant where gall bladder cancer is endemic. Table 1 summarizes the different studies on benign gall bladder pathology and its relation to gall bladder cancer.

In Chile where gallbladder cancer is endemic, the main factor is ancestry. Native populations, 'Mapouches', have high rates of both gall bladder cancer and gallstones (personal communication). To reduce the incidence of gall bladder cancer, the Chilean Government financially supports prophylactic cholecystectomy in patients with gallstones aged between 35 and 49 years. In endemic areas, gallstones in early age (20 years life expectancy or more) seem a rational approach.

Management

There are three possible ways to manage asymptomatic gallstones:

- Expectant management (observation) for all and laparoscopic cholecystectomy when indicated
- Laparoscopic cholecystectomy for all (routine cholecystectomy)
- Laparoscopic cholecystectomy in selected high-risk groups (selective cholecystectomy).

Expectant management (observation)

As most patients (approximately 80%) are going to remain asymptomatic lifelong, this approach is justifiable. However, the drawback of this approach is that as time passes, a greater number of patients will become symptomatic. Few (1% of asymptomatic patients) may present directly with a complication which increases morbidity and mortality. Based on the Markov model decision and cost analysis from India, expectant management should be adopted in the majority, cholecystectomy is recommended only for special indications such as for patients with haemolytic disorders and stones in endemic areas.[33]

At a population level, advocating surgery with its attendant costs and complications, for the 80% of individuals who will remain asymptomatic, seems unjustifiable. Twenty million people in the United States are estimated to have gallstones. Each laparoscopic cholecystectomy costs around $11584. Currently, over 700,000 cholecystectomies are performed annually in the USA, costing $6.2–$6.5 billion.[34] If laparoscopic cholecystectomy was extended to all those with asymptomatic gallstones, the total cost would be exorbitant. Given the limited manpower and resources available, this would also hamper the management of other surgical conditions.

Serious complications such as bile duct injury after laparoscopic cholecystectomy are rare (0.1%-0.5%),[35-37] but considering the high prevalence of gallstone disease, if cholecystectomy was performed for all, a huge number of patients would sustain bile duct injury. This is a very serious complication, usually requiring prolonged treatment and lifelong follow-up, with a mortality risk of 4%–9%[39-41] and significant effects on quality of life.[38] Case series and cohort studies indicate that complications of laparoscopic cholecystectomy, although rare, include severe bleeding (0.11%–1.97%), abscess (0.14%–0.3%) and bowel injury (0.14%–0.35%) besides biloma and bile duct injury.[42,43] There are also financial implications for the health service and also for those patients who are seriously injured at a relatively younger age during the earning phase of life.

The incidence of post-cholecystectomy syndrome is very variable but may occur in up to 47% of patients.[44] Post-cholecystectomy syndrome encompasses symptoms and

Table 1: Benign gall bladder pathology and its relation to gall bladder cancer

Author	Country	Total population studied	Type of benign GB pathology	Number of subjects who had GBC	Length of follow up	RR (95% CI)
Maringhini *et al*., 1987[28]	USA	2583	GS	5	5632 PY	2.8 (0.9–6.6)
Chow *et al*., 1999[29]	Denmark	60,176	GS	42	471,450 PY 8.3 years	3.6 (2.6–4.9)
Lowenfels *et al*., 1985[30]	USA	2530	GS	131	-	Non-Indians 4.4 (2.6–7.3) Indians 20.9 (8.1–54.0)
Nervi *et al*., 1988[31]	Chile	14,768	GS	-	-	7.0 (5.9–8.3)
Kato *et al*., 1989[32]	Japan	327	GS EF	109	-	34.4 (4.51–266.0)
Okamoto *et al*., 1999[16]	Japan	GS - 7985 GB polyp - 10,926	GS GB polyp	19 (6 with GS)	-	10.8 (4.1–28.4)

GS: Gallstone, GB: Gall bladder, CI: Confidence interval, GBC: Gall bladder cancer, RR: Relative risk

signs due to biliary and non-biliary sequelae of laparoscopic cholecystectomy. Non-biliary sequelae may be reflux esophagitis, irritable bowel syndrome, peptic ulceration or chronic pancreatitis. The biliary sequelae may be bile duct injury, biloma, bile salt-induced gastritis or diarrhoea, retained calculi, abscess, dropped calculi, long cystic duct remnant, biliary stenosis, sphincter of Oddi dyskinesia, etc.

Cholecystectomy results in increased enterohepatic circulation leading to altering gut microbiota with the increasing exposure of intestinal mucosa to bile acid. There are reports of increased risk of colorectal cancer in individuals who had undergone cholecystectomy.[45] Although the evidence is not very strong, one should keep all these issues in mind before advocating cholecystectomy for asymptomatic gallstone patients.

Routine laparoscopic cholecystectomy

Laparoscopic cholecystectomy is a safe procedure with a short hospital stay, early resumption of work and lower complication rates than open procedures. Furthermore, in asymptomatic gallstone patients, the operation is easier as there are fewer adhesions and the planes are virgin. Laparoscopic cholecystectomy prevents all future symptoms and complications related to gallstone disease and will be completely preventive for gall bladder cancer.

Prevention of gall bladder cancer is an argument in favour of routine cholecystectomy in asymptomatic gallstone patients in regions with a high incidence of gall bladder cancer including Northern India, Pakistan, South America (Chile and Columbia), East Asia (Japan and Korea) and Eastern Europe (Poland, Czech Republic and Slovakia). However, most studies on the risk of developing gall bladder cancer come from parts of the world where the disease is not endemic. It is, therefore, important to consider the wide variety of aetiology and lack of data from gall bladder cancer endemic regions and question whether it is safe to follow guidelines generated from non-endemic regions. A pragmatic approach in endemic zones is to determine that if life expectancy is more than 20 years or there is a family history of gall bladder cancer, one should consider prophylactic cholecystectomy.

Selective cholecystectomy

Expectant management and routine cholecystectomy for asymptomatic gallstones have certain disadvantages as indicated above. The third way is to be selective in cases of asymptomatic cholelithiasis. Cholecystectomy will be beneficial to those subjects who are going to be symptomatic, have a high chance of developing complications or will develop malignancy in the future. Several studies have been done to risk stratify those subjects at the highest risk of developing complications of malignancy. As mentioned above, there are certain characteristics of patients and stones which are more prone to develop symptoms or complications. Stones of size >3cm, small stones <3mm with a patent cystic duct, floating stones, and the presence of multiple stones are stone-related factors predisposing to the future development of symptoms. Patient-related factors are life expectancy >20 years or age <55 years, female gender, smokers and higher body mass index.[3,8,11,12]

Management strategy

Asymptomatic gallstone disease is best managed with a risk-based approach. Most patients can be safely observed due to the low risk of complications. Routine cholecystectomy is not recommended for all, given its high cost and rare but serious risks, such as bile duct injury and post-cholecystectomy syndrome. In regions with high gall bladder cancer rates, such as parts of India and East Asia, routine surgery may be considered for those with a family history of cancer. Selective cholecystectomy should be reserved for high-risk patients, including those with large stones, multiple stones or a life expectancy >20 years. Expectant management is the most cost-effective strategy, and a patient-centred approach with shared decision-making should guide management. Regular monitoring should be done for symptomatic changes.

There are also some special sub-groups of patients with asymptomatic gallstone which need a separate discussion.

Special Subgroups

Haemolytic disease

In individuals suffering from haemolytic diseases, such as thalassaemia, the occurrence of gallstones is associated with an increased risk of developing complications. Almost 50% develop complications in 3–5 years after diagnosis. Another concern with these patients is that biliary complications of gallstones and veno-occlusive crises may present with similar manifestations and complications of gallstone disease often precipitate veno-occlusive crises.[46] It has been found that emergency surgery in these patients is associated with a significantly higher morbidity, mortality and hospital stay. Considering all these factors, it is imperative to recommend routine cholecystectomy in patients with chronic haemolytic disease when asymptomatic gallstones are present.[46]

Transplant recipients

The mortality of emergency cholecystectomy in transplant recipients on immunosuppression is high. The management of asymptomatic cholelithiasis in transplant recipients varies according to the organ being transplanted.

Cardiac transplant

In patients undergoing cardiac transplant, it has been found that prophylactic post-transplant cholecystectomy is associated with significantly better outcomes as compared to pre-transplant cholecystectomy or expectant management. These findings suggest a role for routine post-transplant cholecystectomy in cardiac transplant recipients.[47]

Renal transplant

Unlike cardiac transplant, prophylactic cholecystectomy in renal transplant patients is not recommended. It has been found that there is no increase in morbidity, mortality or graft loss with expectant management and cholecystectomy only in symptomatic patients.[48]

Bariatric surgery

Bariatric Surgery is associated with substantial weight loss which inherently increases the risk of gallstone formation. The risk of undergoing a cholecystectomy later in life is substantially higher in those undergoing surgery for obesity as compared to the general population. Routine cholecystectomy during bariatric surgery is recommended.[49]

Diabetic patients

Epidemiological studies have demonstrated a higher prevalence of gallstone disease in diabetics. The mechanism of this association is not well understood. Animal models have proposed hepatic insulin resistance to be one of the major factors. As there is a similar course of complications as in symptomatic patients, prophylactic cholecystectomy is not recommended for asymptomatic gallstones in diabetics at present.[50]

Cirrhotic liver disease

Gallstones are twice as common in cirrhotic patients as in the general population. Although laparoscopic cholecystectomy has become the gold standard for symptomatic gallstones, cirrhosis has been considered an absolute or relative contraindication.[51] However, due to the evolution in laparoscopy and advanced perioperative care, surgery for gallstone disease in cirrhotic patients has been performed with acceptable post-operative outcomes.[52] Cirrhotic patients usually remain asymptomatic.

Patients on long-term parenteral nutrition

Patients with short bowel syndrome with long-term parenteral nutrition therapy are at high risk of developing cholelithiasis. The independent predictors are the length of the remnant jejunum and parenteral nutrition dependence. These patients are at high risk of developing complications that should be closely monitored and prophylactic cholecystectomy is recommended.[53]

Perioperative discovery of gallstone during other surgery

These are the group of patients who are found to have gallstones while undergoing another unrelated abdominal surgery. The rate of biliary symptoms and complications is high in these groups of patients after surgery (up to 70%) and 30% need cholecystectomy within a year. Therefore, prophylactic cholecystectomy during index surgery is advisable.[54]

Management

Managing asymptomatic gallstones generally depends on the patient's overall health, age and risk factors such as diabetes, obesity or a family history of gallbladder disease. Clinicians should take time to discuss the pros and cons of cholecystectomy versus observation and allow the patient to reach a joint decision on the preferred treatment.[55] Table 2 summarises the advantages and disadvantages of different management strategies.

Therefore, based on the evidence and arguments presented above, the following pragmatic approach is recommended.

Expectant management

Watchful waiting is a reasonable approach for most patients with asymptomatic gallstones. This involves monitoring the patient's symptoms and periodically assessing the size and number of gallstones with ultrasound imaging. In the absence of symptoms or complications, surgery is usually not recommended. Patients with asymptomatic gallstones should be advised to make lifestyle changes to reduce the risk of developing symptoms or complications. These include maintaining a healthy weight, eating a low-fat diet and exercising regularly.[55]

Selective laparoscopic cholecystectomy

Based on studies and guidelines, the following are conditions where prophylactic cholecystectomy is indicated for asymptomatic gallstone disease.[56]

- Age: Younger patients have more chance of developing symptoms. When life expectancy is more than 20 years, prophylactic cholecystectomy may be considered
- Gender: Women are more likely to develop symptomatic gallstones than men. Therefore, prophylactic cholecystectomy may be considered for asymptomatic women with gallstones, especially those of reproductive age
- Gallstone size: Gallstones larger than 2 cm in diameter are associated with a higher risk of complications. Stones larger than 3 cm are associated with gall bladder cancer in

Table 2: Pros and cons of different management strategies of asymptomatic gallstone

Laparoscopic cholecystectomy		Observation	
Advantages	**Disadvantages**	**Advantages**	**Disadvantages**
Relatively safe surgery	Requires anaesthesia	The majority will have no problem	More patients will be operated in older age when the risk of surgery increases
Short hospital stay usually one episode	Possible complications	No cost for most patients	More adhesions are expected during surgery which may result in more bile duct injury
Early recovery	Risk of bile duct injury, morbidity, and mortality in asymptomatic person	Avoidance of surgery and anaesthesia	Very few may directly present with complications
No risk of future symptoms or complications of gallstone disease	Burden on healthcare system	Majority will have mild symptoms before developing complications	Low but significant risk of gallbladder cancer in endemic areas
The risk of gall bladder cancer eliminated	Less cost-effective		

the long term. Therefore, prophylactic cholecystectomy may be recommended for patients with gallstones larger than 3 cm, especially in endemic areas. Persons with floating calculi and smaller calculi (3mm or less) with a patent cystic duct are at risk of developing pancreatitis. Therefore, laparoscopic cholecystectomy may be considered in these cases

- Gallstone volume: Patients with large volumes of gallstones have a higher risk of developing gall bladder cancer than those with a single gallstone. Therefore, prophylactic cholecystectomy may be considered for such patients
- Gall bladder polyps: Gallstones associated with gallbladder polyps are a separate group and these patients need a decision based on type, size, number and progression of polyps, rather than the presence or type of gallstones.

In low-resource settings, prophylactic laparoscopic cholecystectomy for asymptomatic gallstones raises ethical concerns regarding resource allocation, informed consent and equity. The high cost of the procedure (~$11,584) may divert resources from more urgent healthcare needs, making it financially unsustainable. Informed consent could be compromised, as patients may not fully understand the risks, benefits or long-term consequences. In addition, offering surgery based on demographic factors could exacerbate healthcare disparities. A more ethical approach would involve selective surgery for high-risk patients and conservative management for others, ensuring equitable use of resources.

Conclusion

Only 10%–20% of patients with asymptomatic gallstones will develop symptoms over a 5–10-year period. Biliary colic is usually the first symptom, and it is rare to have severe life-threatening complications without any prior symptoms. Therefore, the current recommendation for managing asymptomatic gallstones is observation and monitoring. This involves regular follow-up visits with a healthcare provider and imaging studies to assess the size and number of stones and to monitor for any changes or development of symptoms. Surgeons and patients should take a joint decision on the risks and benefits of prophylactic cholecystectomy.

In specific high-risk populations, such as patients with large stones (>3 cm), multiple small (<3 mm) stones, associated with common bile duct stone, history of gallbladder disease in a first-degree relative, intraoperative discovery of gallstones in a patient undergoing other surgery, the risk of developing symptoms or complications are higher. In these cases, prophylactic cholecystectomy may be recommended.

Preventive cholecystectomy for asymptomatic gallstones in regions where gallbladder cancer is endemic is a matter of debate. Considering current evidence, prophylactic cholecystectomy should not be routine unless the patient is young (more than 20 years of life expectancy) or has a family history of gall bladder cancer. Conducting a large multicentric randomised controlled prospective clinical trial to manage asymptomatic gallstone would be prudent.

Key points for clinical practice:

- Gallstones are very common in the general population
- Most gallstones will remain asymptomatic
- Complications in most cases are preceded by biliary colic
- A longer asymptomatic period is associated with a lower probability of developing symptoms
- Gallstone is not directly linked to gall bladder cancer in most instances
- Asymptomatic gallstones in gallbladder cancer endemic regions need close monitoring
- The threshold for prophylactic cholecystectomy should be low in gall bladder cancer endemic regions, especially in the younger population.

Authors' contributions

All authors have contributed and approved the manuscript.

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Conflicts of interest

There are no conflicts of interest.

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